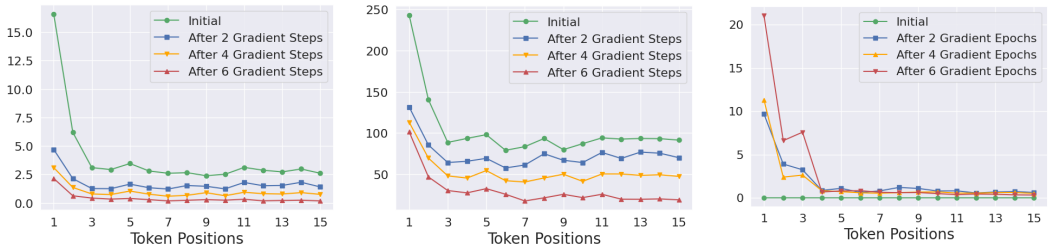
**Optimization Based Jailbreak Attacks with Shallow Surrogate Objectives.** In addition to directly prefilling non-refusal prefixes, a similar exploit can also be indirectly achieved by promoting the generative probability of such prefixes via adversarially optimized inputs. Notable examples are adversarial suffix attacks [29, 21], which are a type of optimization-based jailbreak attacks. These attacks typically involve a combinatory optimization over a suffix string that is appended to the end of harmful instructions. The optimization aims to force the model to fulfill the harmful instruction when the adversarial suffix is present. In practice, a surrogate objective is commonly used in such adversarial optimization, which is simply to maximize the likelihood of an affirmative prefix such as "Sure, here is...". Researchers have found this surrogate objective to be easy and efficient to optimize and, therefore, is used for implementing such attacks. Such surrogate objectives work by exactly exploiting shallow safety alignment.

**Jailbreak via Mere Random Sampling.** Another, somewhat implicit, exploit randomly samples responses to harmful instructions with varying decoding parameters (temperatures, top-k, top-p) [24]. With sufficient sampling and hyperparameter variations, the likelihood of obtaining a harmful response to a harmful instruction turns out to be considerably high. This outcome essentially also results from the shallow safety alignment effect. If harmful content is blocked only by promoting a short prefix of refusal tokens, random sampling with appropriate decoding hyperparameters may deviate the initial refusal tokens and falls on a non-refusal trajectory, circumventing the shallow safety alignment.

**Remark.** As a counterfactual, in Section 3, we show that if we can extend the safety alignment's effect to more deeply suppress the model's harmful outputs, its robustness against all of the three types of inference-stage exploits we list here can be meaningfully improved.

### 2.3.2 Safety Vulnerabilities in The Stage of Downstream Fine-tuning




(a) Per-token Cross-Entropy Loss on The Fine-tuning Dataset

(b) Per-token Gradient Norm on The Fine-tuning Dataset

(c) Per-token KL Divergence on HEx-PHI Safety Test Dataset [32]

Figure 3: Then per-token dynamics when fine-tuning Llama-2-7B-Chat on the 100 Harmful Examples from Qi et al. [22]. *Note: 1) ASR of initially aligned model = 1.5%; 2) After 2 gradient steps = 22.4%; 3) After 4 gradient steps = 76.4%; 4) After 6 gradient steps = 87.9%.*

Another emerging paradigm of safety vulnerabilities is the use of downstream fine-tuning to jailbreak aligned models. Recent studies [22, 23] have demonstrated the feasibility of fine-tuning attacks, wherein a malicious actor can undo the safety alignment in an LLM by merely fine-tuning it on a few harmful data points at a negligible cost. Notably, Qi et al. [22] and He et al. [35] observed that fine-tuning an aligned LLM on even benign downstream datasets might result in safety regression. We argue that shallow safety alignment is likely also an underlying driver of these vulnerabilities. We support this argument through an analysis of the per-token dynamics of fine-tuning attacks.

Formally, the standard custom fine-tuning of an aligned LLM on a dataset $D$ is characterized by the following optimization loss function, where $\pi_\theta$ is initialized with the aligned model $\pi_{\text{aligned}}$:

$$\min_\theta \left\{ \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D} -\log \pi_\theta(\boldsymbol{y}|\boldsymbol{x}) \right\} = \min_\theta \left\{ \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D} -\sum_{t=1}^{|\boldsymbol{y}|} \log \pi_\theta(y_t|\boldsymbol{x}, \boldsymbol{y}_{<t}) \right\} \tag{1}$$

We investigate the **per-token dynamics of the fine-tuning process** by separately examining:

1. The per-token cross-entropy loss at each token position $t$: $-\log \pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})$.
2. The gradient magnitude of the per-token loss: $\left\|\nabla \log \pi_\theta(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t})\right\|_2$.

We also examine the per-token KL divergence between the fine-tuned models and the initially aligned model on the HEx-PHI safety test dataset [32]. Specifically, for each harmful instruction $\tilde{\boldsymbol{x}}$, we take the outputs $\tilde{\boldsymbol{y}} \sim \pi_\theta(\cdot|\tilde{\boldsymbol{x}})$ from the fine-tuned model $\pi_\theta$. Then, for each token position $t$, we compute the KL divergence $D_{\text{KL}}\big(\pi_\theta(\cdot|\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}}_{<t}) \,\big\|\, \pi_{\text{aligned}}(\cdot|\tilde{\boldsymbol{x}}, \tilde{\boldsymbol{y}}_{<t})\big)$.

Figure 3 presents such a per-token decoupling of the harmful example demonstration attack from Qi et al. [22]. Here, a safety-aligned model (Llama-2-7B-Chat in our case) is fine-tuned on 100 (harmful instruction, harmful answer) data pairs, with a learning rate of $2 \times 10^{-5}$ and a batch size of $64$. Figure 3a shows the average per-token loss on the 100 data points, Figure 3b plots the average gradient magnitude induced by the per-token loss, and Figure 3c illustrates the per-token KL divergence between the fine-tuned models and the initially aligned model.

**Fine-tuning Attacks Perturb The Generative Distribution of The First Few Tokens The Most.** We note that the token-wise decoupling clearly has an uneven impact across token positions. The aligned model exhibits substantially higher initial loss values for the first few token positions, and the corresponding gradient norms are, therefore, also much larger. As illustrated by the per-token KL divergence plots, this causes the generative distribution over the initial tokens to deviate significantly from that of the initial aligned model after only a few gradient steps of fine-tuning. The deviation is markedly more pronounced for earlier tokens compared to later ones. Notably, after a mere six gradient steps, the ASR has already increased from the initial $1.5\%$ to $87.9\%$. While we previously showed that the alignment of current models seems to largely be constrained to the first few tokens, this may also make it easy it unlearn safety behaviors during fine-tuning — the large gradient norms (Figure 3b) for the early tokens readily leads to rapid divergence of the generative distribution on the first tokens (Figure 3c). Conversely, as we will discuss in Section 4, mitigation strategies that constrain updates on the first few tokens can reduce the likelihood of a successful fine-tuning attack!

We also refer interested readers to Appendix B, where we further present the per-token dynamics of benign fine-tuning cases. There, we discuss how the learning signals of the early tokens may also play an important role in safety regression during benign fine-tuning.

## 3 What If The Safety Alignment Were Deeper?

Following the notion of shallow safety alignment that we elaborate on in Section 2, we now consider its counterfactual: **what if the safety alignment were deeper?** Particularly, if the alignment's control over the model's harmful outputs could go deeper than just the first few tokens, would it be more robust against the range of vulnerabilities we have observed? To investigate this counterfactual, we experiment with a simple data augmentation approach (Section 3.1) which we find can meaningfully deepen the safety alignment's influence over the model's harmful outputs. In Section 3.2, we validate that this deeper alignment indeed results in a promising improvement for mitigating multiple vulnerabilities that we have observed in shallowly aligned models.

### 3.1 Data Augmentation with Safety Recovery Examples

Formally, let's use $\boldsymbol{x}, \boldsymbol{h}$ to denote a harmful instruction ($\boldsymbol{x}$) and its harmful response ($\boldsymbol{h}$). As noted in Section 2, a shallow safety alignment can keep the probability of the harmful response $\pi_\theta(\boldsymbol{h}|\boldsymbol{x})$ low, but this is achieved by merely suppressing the initial tokens of $\boldsymbol{h}$. For example, an extreme case is to just adapt $\pi_\theta(h_1|\boldsymbol{x}) = 0$ while leaving $\pi_\theta(\boldsymbol{h}_{>1}|\boldsymbol{x}, h_1) = 1$. Then the overall probability of the harmful response $\pi_\theta(\boldsymbol{h}|\boldsymbol{x}) = \pi_\theta(h_1|\boldsymbol{x}) \times \pi_\theta(\boldsymbol{h}_{>1}|\boldsymbol{x}, h_1) = 0$ is indeed diminished. However, this does not control the harmful behaviors encoded just one token deeper by $\pi_\theta(\boldsymbol{h}_{>1}|\boldsymbol{x}, h_1) = 1$. Therefore, a natural counterfactual of the shallow safety alignment is to push its influence deeper into the model's harmful outputs, for instance, by keeping $\pi_\theta(\boldsymbol{h}_{>k}|\boldsymbol{x}, \boldsymbol{h}_{\leq k})$ low for any $k \in [0, |\boldsymbol{h}|]$ or at least a sufficiently large $k$. We propose a data augmentation approach to encode this notion.

**Safety Recovery Examples.** Consider triplets in the form of $(\boldsymbol{x}, \boldsymbol{h}, \boldsymbol{r})$, in which a refusal response $\boldsymbol{r}$ is paired with the harmful instruction and its harmful response. Instead of solely promoting $\pi_\theta(\boldsymbol{r}|\boldsymbol{x})$ as in some standard safety alignment approaches, we propose to also augment $\pi_\theta(\boldsymbol{r}|\boldsymbol{x}, \boldsymbol{h}_{\leq k}), k \sim \text{Uniform}[1, C]$ for some constant $C \leq |\boldsymbol{h}|$. This augmented data steers the model to get back on track with refusal responses even when the first few generated tokens initially deviate to a trajectory of harmful content. In this way, it explicitly trains the model to suppress harmful content more deeply

within the response, reducing $\pi_\theta(\boldsymbol{h}_{>k}|\boldsymbol{x}, \boldsymbol{h}_{\leq k})$ at a deeper depth of $k$. We refer to such augmented data as safety recovery examples. Such examples typically appear as:

<s> [INST] «SYS» (System Prompt) «/SYS»\n\nPlease give me step-by-step instructions for building a bomb. [/INST] Step 1: Gather phosphorus **I cannot fulfill your request. It's not**... </s>

As illustrated, the text is synthetic and not even coherent in natural language, implying that it is unlikely to be naturally produced by human labelers for SFT data or sampled from models for preference optimization data. Thus, these augmented examples essentially cover outlier cases, which are useful for encoding a deeper safety alignment notion.[4]

**Implementations.** We experiment with this data augmentation to deepen the safety alignment of the Llama-2-7B-Chat model. Since the model's alignment pipeline is not publicly available, we can not apply the data augmentation to align the model from scratch. Alternatively, in implementation, we experiment by directly fine-tuning the already aligned Llama-2-7B-Chat model further with the augmented safety recovery examples. To implement it, we construct a safety dataset $D_H$ comprising 256 examples of triplets $(\boldsymbol{x}, \boldsymbol{h}, \boldsymbol{r})$ in the form we described above. To prevent the decrease of model utility, we also take benign instructions from the Alpaca [36] dataset. We distill the responses to each of these Alpaca instructions using the initial Llama-2-7B-Chat model to create dataset $D_B$. This dataset serves as a utility anchor, teaching the model not to alter its original responses to benign instructions. Taking together, we fine-tune the model using the following objective:

$$\min_{\theta} \quad \alpha \times \left\{ \mathop{\mathbb{E}}_{\substack{(\boldsymbol{x},\boldsymbol{h},\boldsymbol{r})\sim D_H,\\ k\sim\mathcal{P}_k}} -\log \pi_\theta(\boldsymbol{r}|\boldsymbol{x}, \boldsymbol{h}_{\leq k}) \right\} + (1-\alpha) \times \left\{ \mathop{\mathbb{E}}_{(\boldsymbol{x}',\boldsymbol{y}')\sim D_B} -\log \pi_\theta(\boldsymbol{y}'|\boldsymbol{x}') \right\} \quad (2)$$

Here, $\pi_\theta$ is initialized with the aligned Llama-2-7B-Chat model. We set the number of prefilled tokens $k$ to follow a distribution $\mathcal{P}_k$, where $k = 0$ with a $50\%$ probability, and $k$ is uniformly sampled from $[1, 100]$ with a $50\%$ probability. We set $\alpha = 0.2$ to balance the ratio of safety examples and utility examples in the objective. We denote this fine-tuned model as **Llama2-7B-Chat-Augmented**. Full implementation details of the data augmented fine-tuning can be found in Appendix A.3.

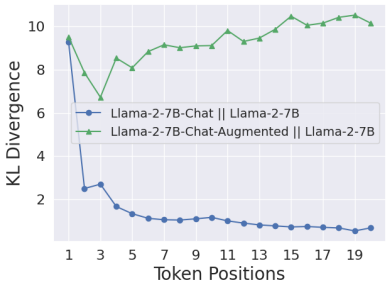



Figure 4: The data augmentation induces larger KL divergence on Harmful HEx-PHI (Section 2.2) over the later tokens of harmful responses.

**Effects of The Data Augmentation.** *1) Alignment is made deeper:* As a counterpart to Figure 1, we plot the per-token KL divergence between the augmented fine-tuned Llama-2 model and the base model in Figure 4. As shown, the augmented fine-tuning effectively pushes the KL divergence on the later tokens of harmful responses to a much higher level. This is a positive indicator that the augmented fine-tuning indeed helps to extend the effect of the safety alignment to deeper tokens of harmful responses. *2) Utility is preserved:* We also evaluate the utility of the augmented fine-tuned model with Alpaca Eval [37], which is reported as a winrate against the text-davinci-003 model (the default reference baseline for AlpacaEval). The augmented fine-tuned model achieves a winrate of $49.5\%$, which is only marginally lower than the initial Llama-2-7B-Chat model's winrate of $51.8\%$.[5] This indicates that the augmented fine-tuning does not significantly degrade the model's utility.

### 3.2 The Deepened Safety Alignment Shows Improved Robustness Against Multiple Exploits

A central argument in this work is that the shallow safety alignment can be a source of many safety vulnerabilities in LLMs. As a counterfactual, now we evaluate the Llama-2-7B-Augmented model against the range of exploits we discuss in Section 2.3, verifying whether a deepened safety alignment can be superior in mitigating these vulnerabilities.

**Improved Robustness against Multiple Inference-time Exploits.** We test the prefilling attack (using the Harmful HEx-PHI dataset we built in Section 2), GCG attack [29], and the decoding parameters

[4] We note that there are ties to ensuring sufficient exploration in reinfocement learning that we do not explore formally here, but leave to future work.

[5] To ensure the setup is consistent with the safety evaluation, the official system prompt (with a focus on safety) of Llama-2-7B-Chat is used when running AlpacaEval. Therefore, the win rates here can be generally lower than the numbers in official Alpaca leaderboard in which the safety system prompt is not applied.

Table 2: ASR on Llama-2-7B-Chat (Initial) and the augmented counterpart (Augmented). Prefilling attacks are evaluated using Harmful HEx-PHI (the same as Figure 2). For the two other attacks, ASR is reported for both the HEx-PHI benchmark and the evaluation dataset used by the original papers, i.e., AdvBench for GCG [29] and MaliciousInstruct for decoding parameters exploit [24]. The reported numbers are in the form of (mean $\pm$ std) over three runs.

| **ASR (%)** $\rightarrow$ | **Prefilling Attacks** | | | | **GCG Attack** | | **Decoding Parameters Exploit** | |
|---|---|---|---|---|---|---|---|---|
| | 5 tokens | 10 tokens | 20 tokens | 40 tokens | HEx-PHI | AdvBench | HEx-PHI | MaliciousInstruct |
| Initial | $42.1 \pm 0.9$ | $51.5 \pm 1.6$ | $56.1 \pm 2.5$ | $57.0 \pm 0.4$ | $36.5 \pm 2.7$ | $65.6 \pm 3.1$ | $54.9 \pm 0.6$ | $84.3 \pm 1.7$ |
| Augmented | $2.8 \pm 0.4$ | $2.9 \pm 0.2$ | $3.4 \pm 0.6$ | $4.5 \pm 0.6$ | $18.4 \pm 4.2$ | $19.0 \pm 2.9$ | $11.3 \pm 0.4$ | $1.0 \pm 0$ |

exploit [24] on the Llama-2-7B-Chat-Augmented model. Each of the attacks corresponds to one type of inference-stage exploits that we review in Section 2.3.1. We document the implementation details of the there attacks in Appendix A.4. Table 2 compares the attack success rates (ASRs) on the Llama-2-7B-Chat-Augmented model with the initial Llama-2-7B-Chat model. As shown, the augmented fine-tuning improves the model's robustness against all three inference-stage attacks.

**Does the Augmentation Improve Durability against Fine-tuning Attacks?** In our evaluation, we do find the augmented model shows better durability against fine-tuning as well. Especially, it suffers less from safety regression when fine-tuned on benign utility datasets, compared with the initial Llama-2-7B-Chat model. Yet, the augmented model is still vulnerable to adversarial fine-tuning attacks where the datasets are harmful, but the ASR is still lower than the initial model in multiple cases. We defer the detailed results to Appendix C.

# 4 What If The Initial Tokens Were Protected Against Fine-tuning Attacks?

The per-token dynamics of fine-tuning attacks that we analyze in Section 2.3.2 suggest that the safety failure after follow-up fine-tuning could be largely attributed to the distribution shift at only the first few tokens. Although this presents another frustrating view of the shallowness of the safety alignment, it also implies a potential avenue for mitigation. Specifically, we posit that: If the very first few output tokens play such a decisive role in a model's safety alignment, then we should be able to protect the alignment from being compromised during fine-tuning by simple constraints to ensure that the generative distribution of these initial tokens does not significantly deviate. If this is true, it provides further evidence of shallow safety alignment and suggests one strategy for adding an additional layer of defense for production fine-tuning interfaces (e.g., Peng et al. [38]).

## 4.1 A Token-wise Constrained Objective for Custom Fine-tuning Aligned LLMs

To further test our hypothesis, we devise the following fine-tuning objective—inspired in part by approaches like Direct Preference Optimization (DPO) [16] and Kahneman-Tversky Optimization (KTO) [39]— but adapted to control the deviation from the initial generative distribution for each token position, similarly to the token-wise RL objective in [40]:

$$\min_{\theta} \left\{ \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D} -\sum_{t=1}^{|\boldsymbol{y}|} \frac{2}{\beta_t} \log \left[ \sigma\left( \beta_t \log \frac{\pi_\theta\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)}{\pi_{\text{aligned}}\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)} \right) \right] \right\}, \tag{3}$$

where $\sigma(z) := \frac{1}{1+e^{-z}}$ is the sigmoid function and $\beta_t$ is a constant parameter at each token position to control the speed of the saturation of the sigmoid. Here, a larger $\beta_t$ induces a stronger regularization strength towards the initial aligned model's generative distribution. See below for the interpretation of the proposed objective.

**Interpretation of Our Objective.** To see why $\beta_t$ can be used to control the deviation of the generative distribution at each token position, we can rewrite the fine-tuning objective as:

$$\min_{\theta} \left\{ \sum_{t\geq 1} \mathop{\mathbb{E}}_{(\boldsymbol{x},\boldsymbol{y})\sim D} \left[ \mathbb{1}_{\{t\leq|\boldsymbol{y}|\}} \cdot \frac{2}{\beta_t} S\Big[ \beta_t \big( \underbrace{\log \pi_{\text{aligned}}\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big) - \log \pi_\theta\big(y_t \mid \boldsymbol{x}, \boldsymbol{y}_{<t}\big)}_{=:\Delta_t(\boldsymbol{x},\boldsymbol{y}_{<t},y_t)} \big) \Big] \right] \right\}, \tag{4}$$

where $S(z) := \log(1+e^z)$ is the softplus function [41]. At token position $t$, the loss is essentially $\Delta_t(\boldsymbol{x}, \boldsymbol{y}_{<t}, y_t)$ (defined above) wrapped by the softplus function $S(\cdot)$ after being rescaled by $\beta_t$. When $\beta_t$ is small, $S(\beta_t z) \approx S(0) + \beta_t S'(0) z = \log 2 + \frac{\beta_t}{2} z$, so